Troubleshooting Network Services

Q1. Simulate a DNS resolution failure on your local machine by editing the hosts file to point www.spotify.com to a wrong IP address, then try to ping the domain and document the error message you receive.

Solution : Simulate DNS resolution failure using the hosts file

Steps:

1. Open Notepad as Administrator.
2. Open this file : C:\Windows\System32\drivers\etc\hosts

3. Add this test entry:

203.0.113.10 www.spotify.com

203.0.113.10 is a reserved example IP address, not Spotify's actual server address.

4. Save the file.
5. Open Command Prompt and run:

ipconfig /flushdns

ping www.spotify.com

Expected observation: Windows may resolve www.spotify.com to 203.0.113.10, and the ping may time out or fail because this is an example address.

Conclusion: An incorrect hosts-file entry can override normal DNS resolution and direct a domain name to the wrong IP address.

Q2. Configure a DHCP server in a network simulator (such as Cisco Packet Tracer or GNS3) and intentionally set an incorrect default gateway; connect a client device and verify the assigned gateway, then describe how you would identify and fix this issue.

Solution : Configure an incorrect DHCP default gateway

Steps in Cisco Packet Tracer:

1. Add one server, one switch, and one PC.
2. Connect them using Copper Straight-Through cables.
3. Configure the server with a static IP, for example 192.168.1.2.
4. Open Server → Services → DHCP and enable DHCP.
5. Configure this example pool:

Setting	**Value**
Start IP Address	192.168.1.10
Subnet Mask	255.255.255.0
Default Gateway	192.168.1.254 (incorrect)
DNS Server	8.8.8.8

6. Set the PC to Desktop → IP Configuration → DHCP.
7. Check the assigned gateway and test connectivity.

Expected observation: The PC may receive an IP address successfully, but its default gateway will be incorrect. It may communicate with devices on its local subnet but fail to reach other networks.

How to fix it:

- Change the DHCP default gateway to the actual router address, such as 192.168.1.1.
- Renew the client's DHCP lease by selecting DHCP again in Packet Tracer.
- Verify the gateway and test connectivity.

Q3. Given a scenario where devices on your simulated network are unable to access websites by name but can by IP address, use the nslookup or dig tool to diagnose the DNS problem and write down the troubleshooting steps and solution.

Solution : Websites work by IP address but not by name

Troubleshooting steps:

Step 1: Test IP connectivity

ping 8.8.8.8

Step 2: Test DNS resolution

nslookup www.google.com

Step 3: Test an alternative DNS server

nslookup www.google.com 8.8.8.8

Step 4: Check DNS configuration

ipconfig /all

Check which DNS servers the computer is using.

Solution: If the configured DNS server is unavailable or incorrect, correct the DNS settings or DHCP DNS option. Flush the local DNS cache if necessary:

ipconfig /flushdns

Q4. A device in your simulated network is not receiving an IP address from the DHCP server and shows an APIPA address (169.x.x.x). List the possible causes for this and document the step-by-step process you would use to resolve it. Hint : Check for network connectivity, DHCP scope exhaustion, and server availability.

Solution : Troubleshoot an APIPA address (169.254.x.x)

An APIPA address is automatically assigned by Windows when it cannot obtain an IPv4 address from a DHCP server.

Possible causes

1. DHCP server is unavailable.
2. Network cable is disconnected or Wi-Fi is not connected.
3. DHCP address pool is exhausted.
4. DHCP service is stopped or incorrectly configured.
5. Firewall or network configuration prevents DHCP communication.

Troubleshooting steps

Step 1: Check the IP address

ipconfig /all

Look for an address beginning with 169.254.

Step 2: Check the local connection

Verify that Wi-Fi is connected or the Ethernet cable is properly plugged in.

Step 3: Renew the IP address

ipconfig /release

ipconfig /renew

Step 4: Check the DHCP server

Verify that the DHCP service is running and that its address pool has available IP addresses.

Step 5: Check the network configuration

Confirm that the DHCP scope, subnet mask, gateway, and VLAN or network connection are correct.

Result: Once the DHCP server is reachable and correctly configured, the client should receive a valid IP address from the DHCP pool.

Q5. Create a table documenting at least three common DNS issues and three common DHCP issues you encountered or researched, including symptoms, possible causes, and recommended solutions for each.

Solution : Common DNS and DHCP issues

Service	Issue	Symptoms	Possible Cause	Recommended Solution
DNS	Incorrect DNS record	Website resolves to the wrong IP	Incorrect A record or hosts entry	Correct the record or remove the incorrect hosts entry
DNS	DNS server unavailable	Domain names fail to resolve	Server outage or network failure	Check DNS server availability and configuration
DNS	DNS cache problem	Old or incorrect resolution	Cached DNS information	Run ipconfig /flushdns
DHCP	APIPA address	Client receives 169.254.x.x	DHCP server unreachable	Check connectivity and DHCP service
DHCP	Scope exhaustion	New clients cannot obtain IP addresses	All available addresses are leased	Expand the scope or release unused leases
DHCP	Incorrect gateway	Local communication works but external networks fail	Wrong DHCP gateway option	Correct the gateway and renew the lease